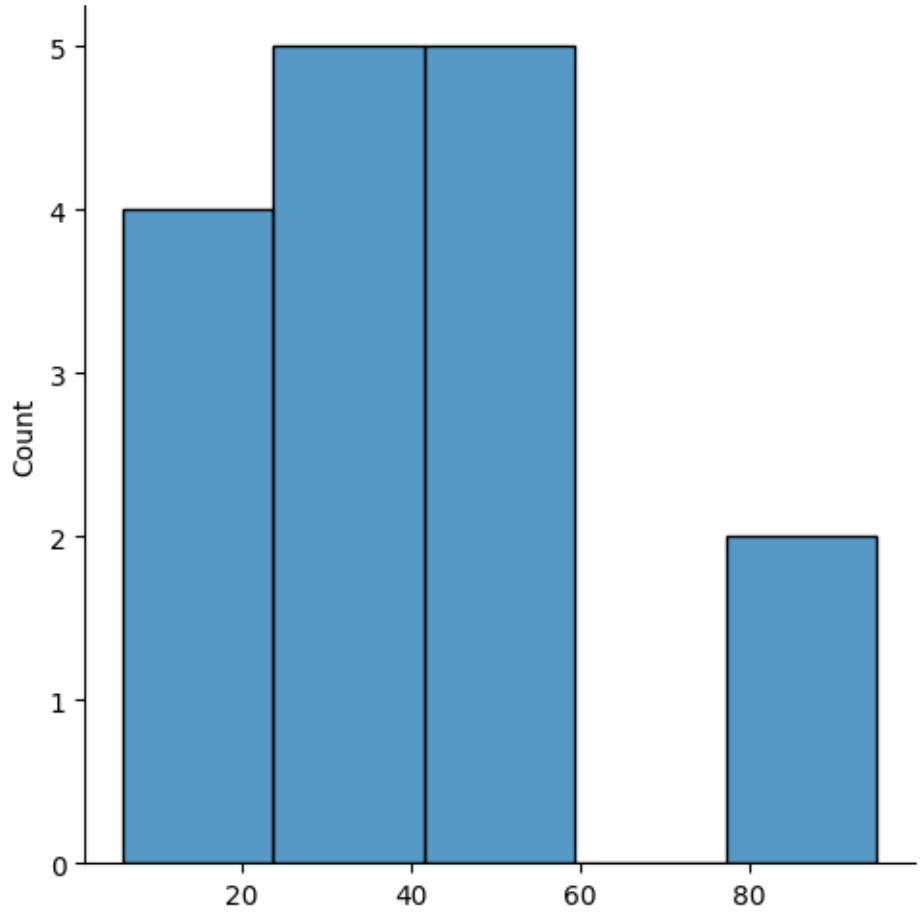
Exp No:12 Experiment to detect outliers in a given data set.

Description: Understand the procedure to identify the outliers in a given dataset

Code:

```
import numpy as np
array=np.random.randint(1,100,16)
array
array([41, 25, 37, 46, 55, 47, 16, 95, 47,  6, 23, 49, 26, 23, 40, 95])
array.mean()
np.float64(41.9375)
np.percentile(array,25)
np.float64(24.5)
np.percentile(array,50)
np.float64(40.5)
np.percentile(array,75)
np.float64(47.5)
np.percentile(array,100)
np.float64(95.0)
def outDetection(array):
 sorted(array)
 Q1,Q3=np.percentile(array,[25,75])
 IQR=Q3-Q1
 lr=Q1-(1.5*IQR)
 ur=Q3+(1.5*IQR)
 return lr,ur
lr,ur=outDetection(array)
lr,ur
(np.float64(-10.0), np.float64(82.0))
import seaborn as sns
%matplotlib inline
sns.displot(array)
```

<seaborn.axisgrid.FacetGrid at 0x7e4064b00ad0>



sns.distplot(array)

/tmp/ipykernel_820/1133588802.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0

Please adapt your code to use either `displot` (a figure-level function with

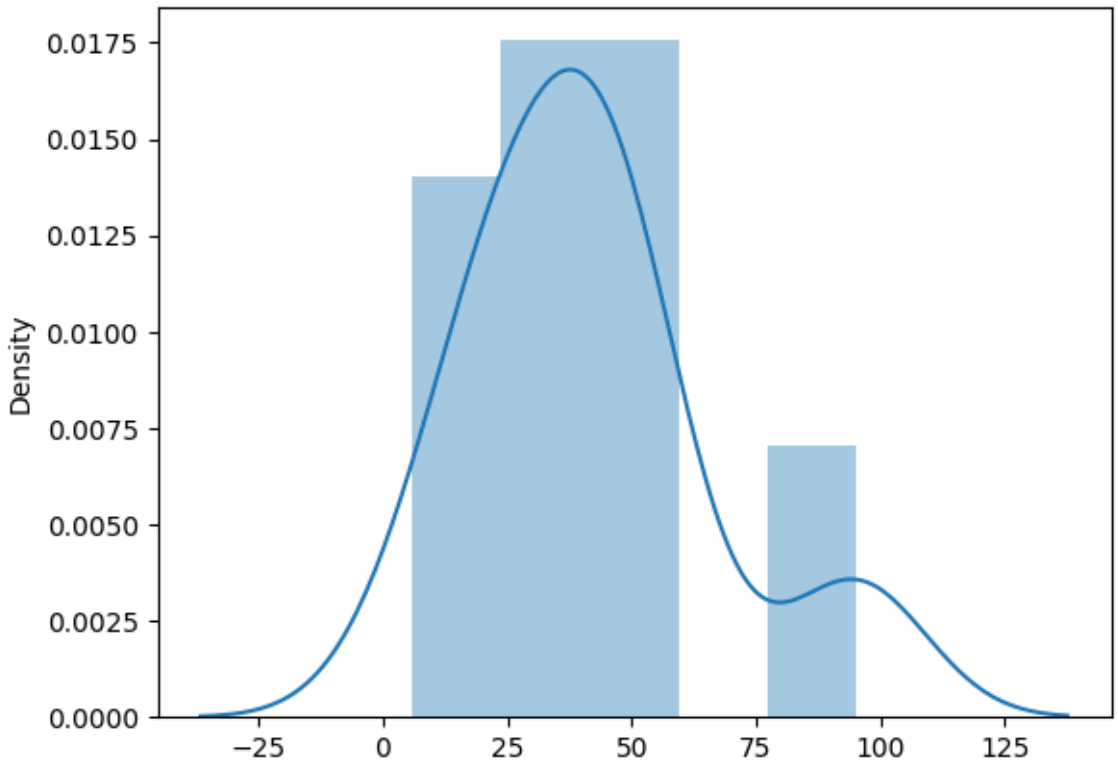
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see

https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

sns.distplot(array)

<Axes: ylabel='Density'>



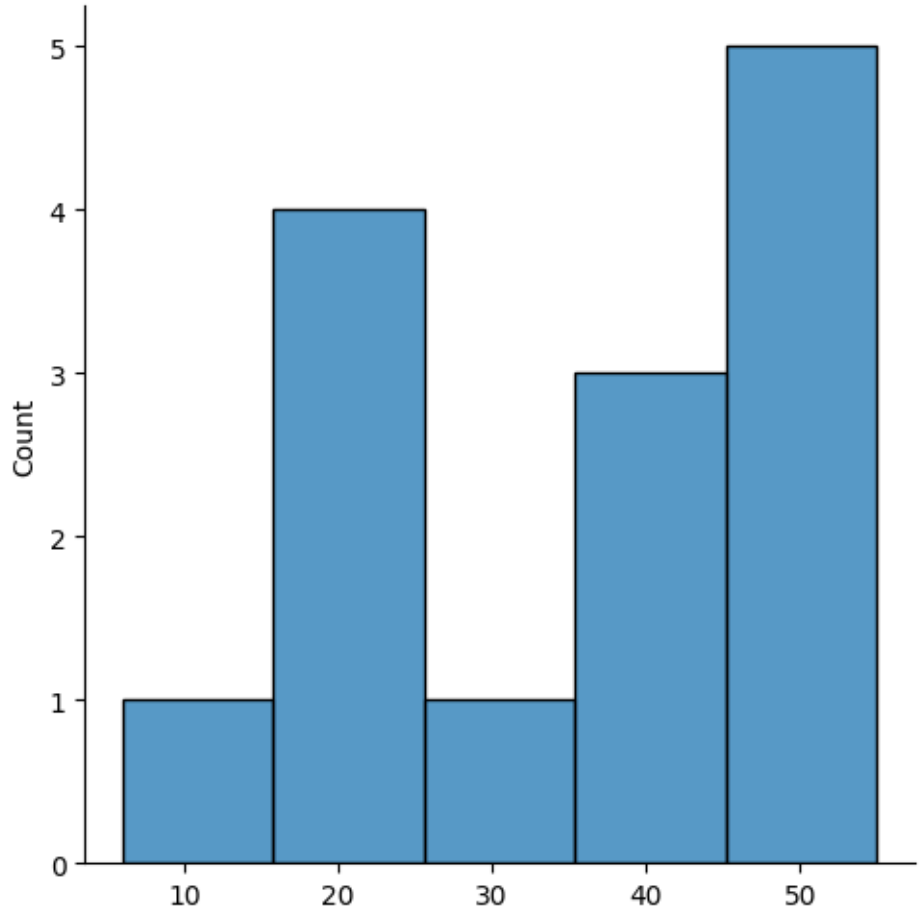
new_array=array[(array>lr) & (array<ur)]

new_array

array([41, 25, 37, 46, 55, 47, 16, 47, 6, 23, 49, 26, 23, 40])

sns.displot(new_array)

<seaborn.axisgrid.FacetGrid at 0x7e406494f890>



lr1,ur1=outDetection(new_array)

lr1,ur1

(np.float64(-11.375), np.float64(81.625))

final_array=new_array[(new_array>lr1) & (new_array<ur1)]

final_array

array([41, 25, 37, 46, 55, 47, 16, 47, 6, 23, 49, 26, 23, 40])

sns.distplot(final_array)

/tmp/ipykernel_820/209491988.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with

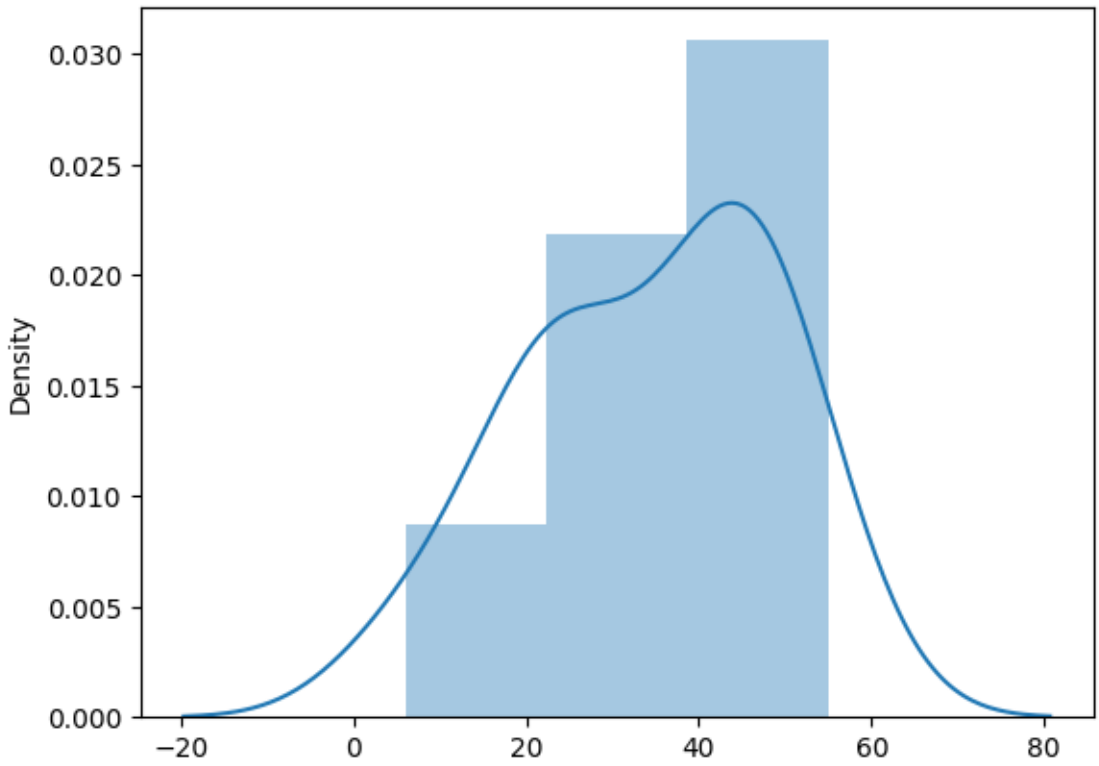
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see

https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

 sns.distplot(final_array)

<Axes: ylabel='Density'>



Result:

Thus, Unserstood the procedure to identify the outliers in a given dataset